

Technical Document #37: Database Systems - Comprehensive Overview

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Database indexing improves query performance by creating data structures that allow faster data retrieval.

Database migration tools manage schema changes and data transformations. Flyway and Liquibase are popular database migration tools.

Relational databases organize data into tables with rows and columns. SQL is used to query and manipulate structured data.

Database replication copies data across multiple servers. It improves availability and provides fault tolerance.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Database replication copies data across multiple servers.
- Database monitoring tracks performance metrics and identifies bottlenecks.
- Database migration tools manage schema changes and data transformations.